Chapter 3 – Vectors

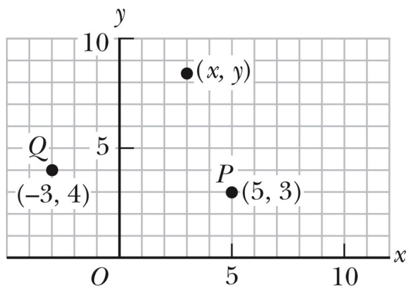
Coordinate systems
Vector arithmetic
Unit vectors
Vectors and scalars
Components
Multiplying vectors

Coordinate System

- Physics often requires a description of a location in space
  - e.g. an objects motion as in previous chapter 2.
- Use various ways to represent this location
  - Cartesian or Rectangular coordinates
  - Plane polar coordinates
- Need to apply trigonometric functions

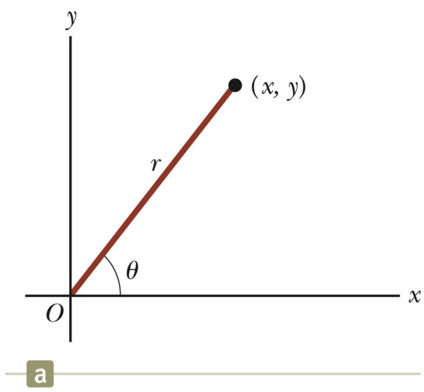
Cartesian or Rectangular Coordinate System

- x– and y– axes intersect at the origin
- Points are labeled (x,y)



Polar Coordinate System

- Origin and reference line are noted
- Point is distance r from the origin in the direction of angle θ, counter clockwise from reference line
  - The reference line is often the x–axis.
- Points are labeled (r,θ)

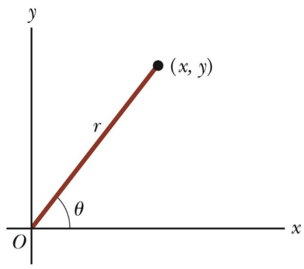


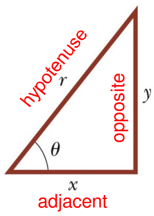
Polar ↔ Cartesian Coordinates

Based on the corresponding right triangle from r and θ

$x = r \cos \theta$

$y = r \sin \theta$





If the Cartesian coordinates are known:

$$\tan\theta = \frac{opp}{adj}$$

$$r = \sqrt{x^2 + y^2}$$

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

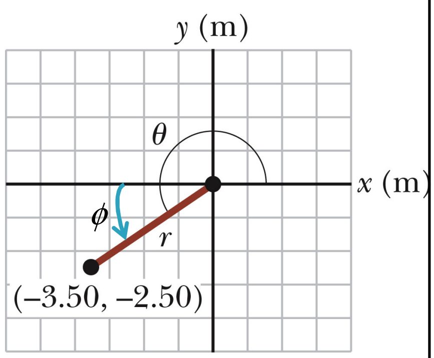
$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$$

Example 3.1

- Work through solution
- The Cartesian coordinates of a point in the *xy* plane are (*x*,*y*)=(–3.50, –2.50)m, as shown in the figure. Find the polar coordinates of this point.

- Repeat with the reference line along the $-x$–axis



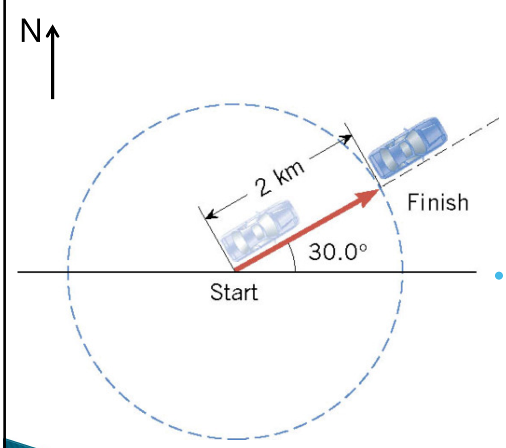
ϕ = 35.5° below the $-x$-axis

Scalars and Vectors

- A *scalar* quantity is completely described by a single value with an appropriate unit and has no direction.
  - temperature, speed, mass
- A *vector* is a physical quantity and is completely specified by a number with an appropriate unit (magnitude) plus a direction:
  - velocity, force, displacement
  - $\overline{F}$ or $\vec{F}$ or $\boldsymbol{F}$ or $\vec{\boldsymbol{F}}$ in texts
  - Handwritten: $\vec{\mathrm{F}}$
  - Magnitude of the vector: $|\vec{F}|$

$$\therefore F = |\vec{F}|$$

Scalars and Vectors

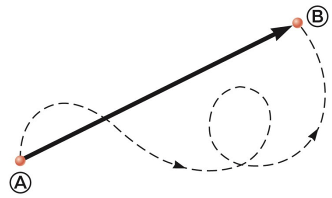


- The car moved a distance of 2 km in a straight line.

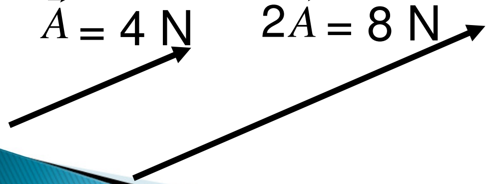
- The car moved a distance of 2 km in a direction 30° north of east.

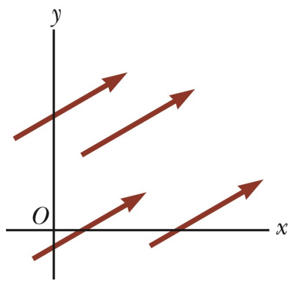
Vectors

- Arrows are used to represent vectors. The direction of the arrow gives the direction of the vector.
- By convention, the length of a vector arrow is proportional to the magnitude of the vector.



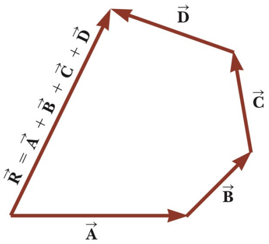
$\vec{A} = 4$ N $\quad 2\vec{A} = 8$ N

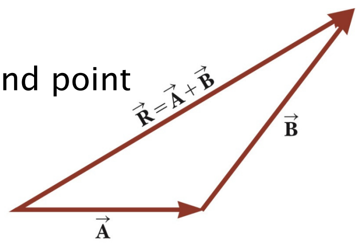




Vector addition–Graphically

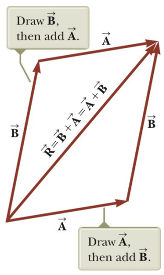
- A particle moves from point 1 to point 2, $\vec{A}$ then from point 2 to point 3, $\vec{B}$
- Draw first vector to scale, then the second vector at the tip of first.
- "Head to tail" method
- Resultant $\vec{R}$ = Starting point to End point
- More than 2 vectors

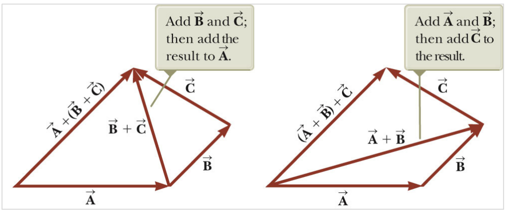




Vector addition–Graphically

- Commutative law
- Associative law

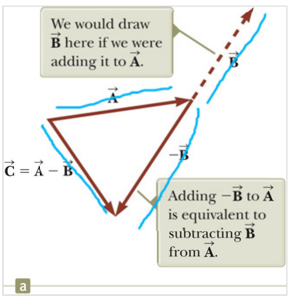




$$\vec{\mathbf{A}} + \left(\vec{\mathbf{B}} + \vec{\mathbf{C}}\right) = \left(\vec{\mathbf{A}} + \vec{\mathbf{B}}\right) + \vec{\mathbf{C}}$$

Vector addition–Geometric

- Vector subtraction $\vec{\mathbf{A}} + \left(-\vec{\mathbf{B}}\right) = \vec{\mathbf{C}} = \vec{\mathbf{A}} - \vec{\mathbf{B}}$
  (Negative vector)



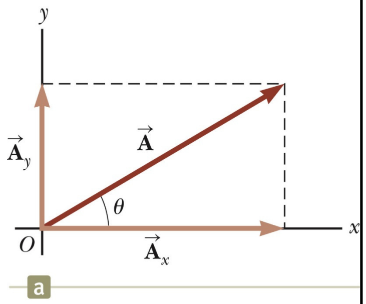
Example

- The magnitudes of displacements $\vec{a}$ and $\vec{b}$ are 3m and 4 m respectively, and $\vec{c} = \vec{a} + \vec{b}$ Considering the various orientations of $\vec{a}$ and $\vec{b}$, what is the (i) maximum and (ii) minimum possible magnitude for $\vec{c}$?

Also go through Example 3.2

Components of vectors

- Graphical addition of vectors – inefficient
- A component of a vector is a projection of the vector on an axis.
- $\vec{A}_x$ is the component of vector $\vec{A}$ along the x-axis, and $\vec{A}_y$ is the component of vector along the y-axis.
- Draw $\perp$ lines to axes.
- Resolve the vector.
- A_x, A_y both positive <u>in this case</u>.
- Generally, 3 components x-, y- and z-components.

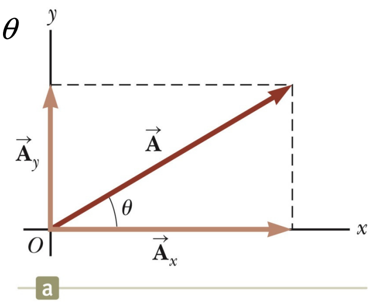


Components of vectors

- First select a set of axes (coordinate system)
- Usually:

$$\cos\theta = \frac{adj}{hyp} \qquad \text{then } A_x = A\cos\theta$$

$$\sin\theta = \frac{opp}{hyp} \qquad \text{then } A_y = A\sin\theta$$



- $A = \sqrt{A_x^2 + A_y^2}$

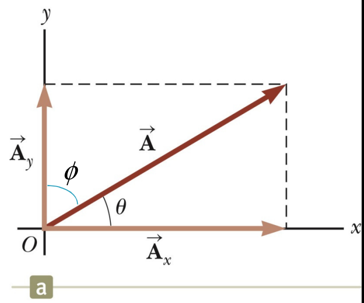
$$\tan\theta = \frac{opp}{adj} = \frac{A_y}{A_x}$$

Components of vectors

- Now if you use the other angle – ϕ:

$$\cos\phi = \frac{adj}{hyp} \qquad \text{then } A_y = A\cos\phi$$

$$\sin\phi = \frac{opp}{hyp} \qquad \text{then } A_x = A\sin\phi$$



- $A = \sqrt{A_x^2 + A_y^2}$

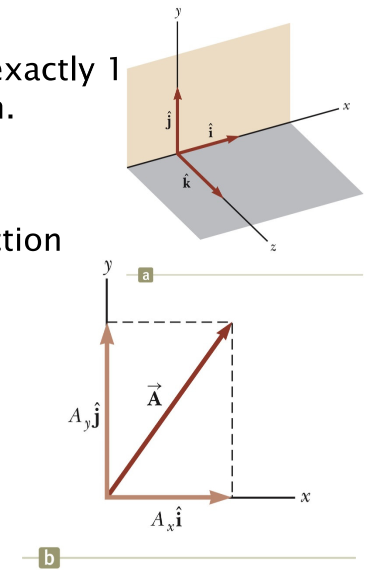
$$\tan\phi = \frac{A_x}{A_y}$$

Example

- A small plane leaves an airport on an overcast day and is later sighted 215 km away in a direction making an angle of 22° east of due north. How far east and how far north is the plane from the airport when sighted?

Unit Vectors

- A unit vector has a magnitude of exactly 1 and points in a particular direction.
- Dimensionless and unitless
- Purpose is to specify direction
- Labelled: $\hat{\mathbf{i}}, \hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$ for x, y, z direction
  - Right handed co-ordinate system.
    $\vec{A} = A_x\,\hat{\mathrm{i}} + A_y\,\hat{\mathrm{j}}$
  - Vector components (units)
    - $A_x\,\hat{\mathrm{i}}$ and $A_y\,\hat{\mathrm{j}}$
  - Scalar components (units)
    - A_x and A_y



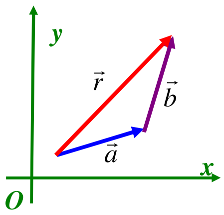
Adding Vectors by Components

- Consider the vectors

$$\vec{a} = a_x\,\hat{\mathrm{i}} + a_y\,\hat{\mathrm{j}} + a_z\,\hat{\mathrm{k}}$$

$$\vec{b} = b_x\,\hat{\mathrm{i}} + b_y\,\hat{\mathrm{j}} + b_z\,\hat{\mathrm{k}}$$

$$\vec{r} = \vec{a} + \vec{b}$$



- Then

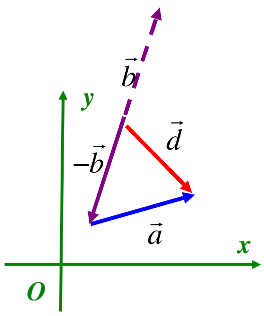
$$r_x = a_x + b_x$$

$$r_y = a_y + b_y$$

$$r_z = a_z + b_z$$

$$\vec{r} = r_x\hat{i} + r_y\hat{j} + r_z\hat{k}$$

Subtraction of Vectors



- Consider the vectors

$$\vec{a} = a_x\,\hat{\mathrm{i}} + a_y\,\hat{\mathrm{j}} + a_z\,\hat{\mathrm{k}}$$

$$\vec{b} = b_x\,\hat{\mathrm{i}} + b_y\,\hat{\mathrm{j}} + b_z\,\hat{\mathrm{k}}$$

$$\vec{d} = \vec{a} - \vec{b}$$

- Then

$$d_x = a_x - b_x$$

$$d_y = a_y - b_y$$

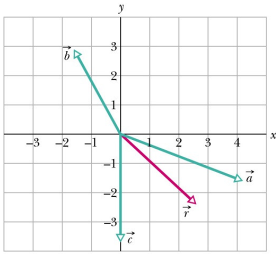
$$d_z = a_z - b_z$$

- and

$$\vec{d} = d_x\,\hat{\mathrm{i}} + d_y\,\hat{\mathrm{j}} + d_z\,\hat{\mathrm{k}}$$

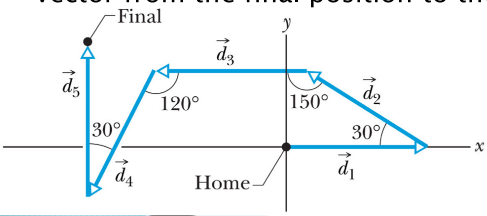
Example

- Consider the three vectors as shown in the figure.
  $$\vec{a} = (4.2\ \text{m})\hat{\text{i}} - (1.5\ \text{m})\hat{\text{j}}$$
  $$\vec{b} = (-1.6\ \text{m})\hat{\text{i}} + (2.9\ \text{m})\hat{\text{j}}$$
  $$\vec{c} = (-3.7\ \text{m})\hat{\text{j}}$$
  What is their vector sum (also shown in red)?



Example

- When an ant wants to return to its home next, it effectively sums its displacements along the axes of the system to calculate a vector that points directly home. Consider an ant making five runs of 6.0 cm each as shown in figure.
  - What are the magnitude and direction of the ants net displacement vector?
  - What are the magnitude and direction of the homeward vector from the final position to the nest?



Example

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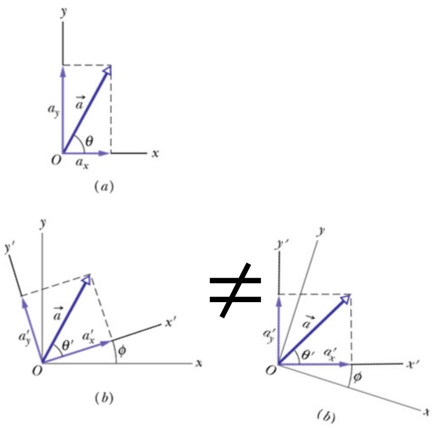
  - What are the magnitude and direction of the ants net displacement vector?

    $\vec{r} = -8.3\ \text{cm}\,\hat{\text{i}} + 3.8\ \text{cm}\,\hat{\text{j}}$
  - What are the magnitude and direction of the homeward vector from the final position to the nest?

    $r_{\text{home}} = +8.3\ \text{cm}\,\hat{\text{i}} - 3.8\ \text{cm}\,\hat{\text{j}}$

Vectors and Physics

- Usually have shown axes for vectors parallel to the page edges.
- No particular reason for this.
- Can rotate the axis and still have the “same” resultant vector
- Use this a lot in Physics particularly Mechanics.
- DO NOT (!) rotate the vectors!!

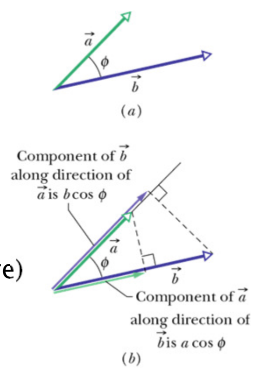


Multiplying Vectors

- Strauss Chapter 4
- Multiplying a vector by a scalar quantity S (Strauss Ch 4.2
  - New vector with S time the original vector's magnitude
  - Direction is same as original vector if S is positive
  - Divide vector by S = multiply by $1/S$.
- Multiplying a vector by a vector
  - Two methods (Strauss Ch 4.3 and 4.4)
  - One method produces a scalar – scalar (or dot) product
  - One method produces another vector – vector (or cross) product

Scalar product (Ch 7)

- Two vectors $\vec{a}$ and $\vec{b}$
- Scalar product or dot product
  $\vec{a} \cdot \vec{b} = ab\cos\phi$
  - a, b magnitudes of the vectors
  - ϕ – angle between the directions of two vectors
- The product of the magnitude of one vector and the scalar component of the other vector along the direction of the first. (see (b) in figure)
- If ϕ = 0° then dot product is a max.
- If ϕ = 90° then dot product is 0.
- $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$
- $\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z$



Scalar Product

- Scalar product using unit vector notation

$$
\begin{aligned}
\vec{a}\cdot\vec{b} &= \left(a_x\hat{i}+a_y\hat{j}+a_z\hat{k}\right)\cdot\left(b_x\hat{i}+b_y\hat{j}+b_z\hat{k}\right)\\
&= \left(a_x\hat{i}\cdot b_x\hat{i}\right)+\left(a_x\hat{i}\cdot b_y\hat{j}\right)+\left(a_x\hat{i}\cdot b_z\hat{k}\right)\\
&\quad+\left(a_y\hat{j}\cdot b_x\hat{i}\right)+\left(a_y\hat{j}\cdot b_y\hat{j}\right)+\left(a_y\hat{j}\cdot b_z\hat{k}\right)\\
&\quad+\left(a_z\hat{k}\cdot b_x\hat{i}\right)+\left(a_z\hat{k}\cdot b_y\hat{j}\right)+\left(a_z\hat{k}\cdot b_z\hat{k}\right)\\
&= a_xb_x+a_yb_y+a_zb_z
\end{aligned}
$$

Example

- Vectors $\vec{C}$ and $\vec{D}$ have magnitudes of 3 units and 4 units, respectively. What is the angle between the directions of the vectors if the dot product equals (i) 0, (ii) 12 units, and (iii) −12 units?

Examples

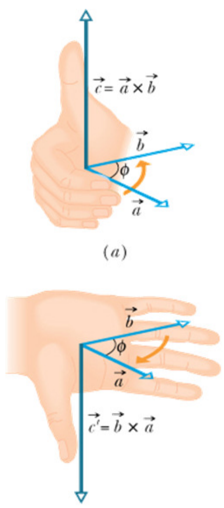
- What is the angle ϕ between $\vec{a} = 3.0\hat{i} - 4.0\hat{j}$ and $\vec{b} = -2.0\hat{i} + 3.0\hat{k}$?

Vector Product (Ch11)

- Two vectors $\vec{a}$ and $\vec{b}$
- Vector product or cross product

$$c = \left|\vec{a} \times \vec{b}\right| = ab\sin\phi$$

  - c – Magnitude of the third vector
  - a, b magnitudes of two initial vectors
  - ϕ – smaller of the two angle between the two vectors.
- If ϕ = 0° then cross product is 0.
- If ϕ = 90° then cross product is a max.
- Direction of $\vec{c}$ is $\perp$ to plane containing $\vec{a}$ and $\vec{b}$
- RIGHT-hand rule:



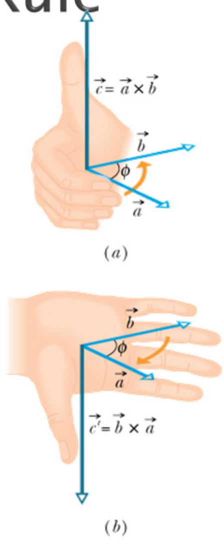
(a)

(b)

Vector Product – RH Rule

- $\vec{c} = \vec{a} \times \vec{b}$
- Place the vectors tail to tail.
- Point the fingers of your right hand in the direction of the first vector.
- your fingers, through the smallest angle between $\vec{a}$ and $\vec{b}$, from vector $\vec{a}$ to $\vec{b}$.
- Your outstretched thumb points in the direction of $\vec{c}$.
- Commutative law does NOT apply.

  $\vec{a} \times \vec{b} = -\left(\vec{b} \times \vec{a}\right)$



Vector Product

- $\vec{c} = \vec{a} \times \vec{b}$
- $\vec{a} \times \vec{b} = \left(a_x\hat{i} + a_y\hat{j} + a_z\hat{k}\right) \times \left(b_x\hat{i} + b_y\hat{j} + b_z\hat{k}\right)$

$$= \cancel{\left(a_x\hat{i} \times b_x\hat{i}\right)} + \left(a_x\hat{i} \times b_y\hat{j}\right) + \left(a_x\hat{i} \times b_z\hat{k}\right)$$

$$+ \left(a_y\hat{j} \times b_x\hat{i}\right) + \cancel{\left(a_y\hat{j} \times b_y\hat{j}\right)} + \left(a_y\hat{j} \times b_z\hat{k}\right)$$

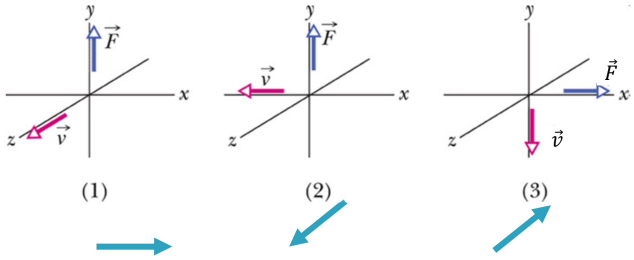
$$+ \left(a_z\hat{k} \times b_x\hat{i}\right) + \left(a_z\hat{k} \times b_y\hat{j}\right) + \cancel{\left(a_z\hat{k} \times b_z\hat{k}\right)}$$

$$= \left(a_y b_z - a_z b_y\right)\hat{i} + \left(a_z b_x - a_x b_z\right)\hat{j} + \left(a_x b_y - a_y b_x\right)\hat{k}$$

- Determinant method – Strauss ch 4

Example

- If $\vec{F} = q\left(\vec{v}\times\vec{B}\right)$ and $\vec{v}$ is perpenducular to $\vec{B}$, then what is the direction of $\vec{B}$ in each of the three situations below:



Example

- The figure shows vector $\vec{a}$ lying in the xy plane, with a magnitude of 18 units and pointing in the direction of 250° from the $+x$–direction. Also, vector $\vec{b}$ has a magnitude of 12 units and points along the $+z$ direction. What is the vector product $\vec{c} = \vec{a}\times\vec{b}$?